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3.2.7. Student Data Analysis and Operations

05:38

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- **Print all student details:** Display the complete details of all students, including roll numbers and marks for all subjects.
- **Find total students:** Determine the total number of students in the dataset.
- **Print all student roll numbers:** Extract and print the roll numbers of all students.
- **Print Subject 1 marks:** Extract and print the marks of all students in Subject 1.
- **Find minimum marks in Subject 2:** Identify the lowest marks in Subject 2.
- **Find maximum marks in Subject 3:** Identify the highest marks in Subject 3.
- **Print all subject marks:** Display the marks of all students for each subject.
- **Find total marks of students:** Compute the total marks for each student across all subjects.
- **Find the average marks of each student:** Compute the average marks for each student.
- **Find average marks of each subject:** Compute the average marks for all students in each subject.
- **Find average marks of Subject 1 and Subject 2:** Compute the average marks for Subject 1 and Subject 2.
- **Find average marks of Subject 1 and Subject 3:** Compute the average marks for Subject 1 and Subject 3.
- **Find the roll number of the student with maximum marks in Subject 3:** Identify the student with the highest marks in Subject 3 and print their roll number.
- **Find the roll number of the student with minimum marks in Subject 2:** Identify the student with the lowest marks in Subject 2 and print their roll number.
- **Find the roll number of students who scored 24 marks in Subject 2:** Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.

Operation...

Submit

9 print("Total Students:", a.shape[0])

10

11 # 3. Print all student Roll numbers

12 print("All Student Roll Nos", a[:,0])

13

14 # 4. Print subject 1 marks

15 print("Subject 1 Marks", a[:,1])

16

17 # 5. print minimum marks of Subject 2

18 print("Min marks in Subject 2", np.min(a[:, 2]))

Average time 0.023 s 23.00 ms

Maximum time 0.023 s 23.00 ms

1 out of 1 shown test case(s) passed

Test case 1 23 ms Debug

Expected output

Actual output

[[301, 67, 77, 88]]

[[301, 67, 77, 88]]

[[302, 78, 88, 77]]

[[302, 78, 88, 77]]

[[303, 45, 56, 89]]

[[303, 45, 56, 89]]

[[304, 88, 98, 45]]

[[304, 88, 98, 45]]

[[305, 78, 88, 99]]

[[305, 78, 88, 99]]

[[306, 24, 78, 30]]

[[306, 24, 78, 30]]

Terminal Test cases

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3.2.6. Numpy: Searching, Sorting, Counting, Broadcasting

08:35

The given code in the editor takes a single array, `array1`, as space-separated integers as input from the user.

Additionally, it takes the following inputs:

- `search_value`: The value to search for in the array.
- `count_value`: The value to count its occurrences in the array.
- `broadcast_value`: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:

- Searching**: Find the indices where `search_value` appears in `array1` and print these indices.
- Counting**: Count how many times `count_value` appears in `array1` and print the count.
- Broadcasting**: Add `broadcast_value` to each element of `array1` using broadcasting, and print the resulting array.
- Sorting**: Sort `array1` in ascending order and print the sorted array.

**Input Format:**

- A single line containing space-separated integers representing `array1`.
- An integer `search_value` represents the value to search for in the array.
- An integer `count_value` represents the value to count in the array.
- An integer `broadcast_value` represents the value to add to each element of the array.

**Output Format:**

- The indices where `search_value` occurs in `array1`.
- The count of occurrences of `count_value` in `array1`.

arrayOpe...

Submit

1 import numpy as np

2

3 # Input array from the user

4 array1 = np.array(list(map(int, input().split())))

5

6 # Searching

7 search\_value = int(input("Value to search: "))

8 count\_value = int(input("Value to count: "))

9 broadcast\_value = int(input("Value to add: "))

10

Average time 0.017 s 17.00 ms

Maximum time 0.019 s 19.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 10 ms Debug

Expected output

1 1 1 2 2 2

Value to search: 1

Value to count: 2

Value to add: 2

[0 1 2]

3

Actual output

1 1 1 2 2 2

Value to search: 1

Value to count: 2

Value to add: 2

[0 1 2]

3

Terminal Test cases

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3.2.5. Numpy: Copying and Viewing Arrays

04:59

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the `original_array` and assigning it to `view_array`.
- Creating a copy of the `original_array` and assigning it to `copy_array`.

After completing these steps, observe how modifying the view affects the `original_array`, while modifying the copy does not.

**Input Format:**

- A single line of space-separated integers.

**Output Format:**

- After modifying the view:

Original array after modifying view: `<original_array>`  
View array: `<view_array>`

- After modifying the copy:

Original array after modifying copy: `<original_array>`  
Copy array: `<copy_array>`

Sample Test Cases

copyAnd...

Submit

Debugger

```
4
5 # Original array
6 original_array = np.array(inputlist)
7
8 # Create a view
9 view_array = original_array.view()
10
11 # Create a copy
12 copy_array = original_array.copy()
13
```

Average time  
0.007 s  
7.00 ms

Maximum time  
0.011 s  
11.00 ms

2 out of 2 shown test case(s) passed  
2 out of 2 hidden test case(s) passed

Test case 1 11 ms Debug

Expected output  
10 20 30 40 50 60 70 80

Actual output  
10 20 30 40 50 60 70 80

Original array after modifying view: [99 20 30 40 50 60 70 80]

Original array after modifying view: [99 20 30 40 50 60 70 80]

View array: [99 20 30 40 50 60 70 80]

View array: [99 20 30 40 50 60 70 80]

Original array after modifying copy: [99 20 30 40 50 60 70 80]

Original array after modifying copy: [99 20 30 40 50 60 70 80]

Copy array: [10 88 30 40 50 60 70 80]

Copy array: [10 88 30 40 50 60 70 80]

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3.2.5. Numpy: Copying and Viewing Arrays

04:59

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the `original_array` and assigning it to `view_array`.
- Creating a copy of the `original_array` and assigning it to `copy_array`.

After completing these steps, observe how modifying the view affects the `original_array`, while modifying the copy does not.

**Input Format:**

- A single line of space-separated integers.

**Output Format:**

- After modifying the view:

Original array after modifying view: <original\_array>

View array: <view\_array>

- After modifying the copy:

Original array after modifying copy: <original\_array>

Copy array: <copy\_array>

Sample Test Cases

copyAnd...

Submit

Debugger

```
4
5 # Original array
6 original_array = np.array(inputlist)
7
8 # Create a view
9 view_array = original_array.view()
10
11 # Create a copy
12 copy_array = original_array.copy()
13
```

Average time

0.007 s

7.00 ms

Maximum time

0.011 s

11.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

11 ms

Debug

Expected output

Actual output

10 20 30 40 50 60 70 80

10 20 30 40 50 60 70 80

Original array after modifying view: [99 20 30 40 50 60 70 80]

Original array after modifying view: [99 20 30 40 50 60 70 80]

View array: [99 20 30 40 50 60 70 80]

View array: [99 20 30 40 50 60 70 80]

Original array after modifying copy: [99 20 30 40 50 60 70 80]

Original array after modifying copy: [99 20 30 40 50 60 70 80]

Copy array: [10 88 30 40 50 60 70 80]

Copy array: [10 88 30 40 50 60 70 80]

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3.2.4. Numpy: Arithmetic and Statistical Operations, Mathematical Operations, Bitw... 09:14

You are given two arrays A and B. Your task is to complete the function array\_operations, which will convert these lists into NumPy arrays and perform the following operations:

1. Arithmetic Operations:

• Compute the element-wise sum, difference, and product of the two arrays.

2. Statistical Operations:

• Calculate the mean, median, and standard deviation of array A.

3. Bitwise Operations:

• Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex:  $A_1$  OR  $B_1$ ).

Input Format:

• The first line contains space-separated integers representing the elements of array A.

• The second line contains space-separated integers representing the elements of array B.

Output Format:

• For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

Sample Test Cases

different...

1 import numpy as np

2

3 def array\_operations(A, B):

4

5     # Convert A and B to NumPy arrays

6     arr\_A=np.array(A)

7     arr\_B=np.array(B)

8     # Arithmetic Operations

9     sum\_result = arr\_A +arr\_B

10    diff\_result = arr\_A - arr\_B

Average time

0.018 s

18.33 ms

Maximum time

0.029 s

29.00 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 29 ms

Debug

Expected output

1 2 3 4

5 6 7 8

Element-wise Sum: 6 8 10 12

Element-wise Difference: -4 -4 -4 -4

Element-wise Product: 5 12 21 32

Mean of A: 2.5

Actual output

1 2 3 4

5 6 7 8

Element-wise Sum: 6 8 10 12

Element-wise Difference: -4 -4 -4 -4

Element-wise Product: 5 12 21 32

Mean of A: 2.5

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Test cases

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3.2.3. Numpy: Custom Sequence Generation

03:04

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using `numpy` based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

- The program should print the generated sequence based on the input values.

Sample Test Cases

customS...

Submit

1 `import numpy as np`

2

3 `# Take user input for the start, stop, and step of the sequence`

4 `start = int(input())`

5 `stop = int(input())`

6 `step = int(input())`

7

8 `# Generate the sequence using np.arange()`

9 `sequence = np.arange(start, stop, step)`

10 `# Print the generated sequence`

Average time

0.010 s

9.75 ms

Maximum time

0.013 s

13.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 13 ms

Debug

Expected output

3

10

2

[3 5 7 9]

Actual output

3

10

2

[3 5 7 9]

Test case 2 10 ms

Terminal

Test cases

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3.2.2. Numpy: Horizontal and Vertical Stacking of Arrays

17:11

You are given two arrays `arr1` and `arr2`. You need to perform horizontal and vertical stacking operations on them using NumPy.

- Horizontal Stacking:** Stack the two matrices horizontally (side by side).
- Vertical Stacking:** Stack the two matrices vertically (one below the other).

**Input Format:**

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

**Output Format:**

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

Sample Test Cases

stacking.py

Submit

1 import numpy as np

2

3 # Input matrices

4 print("Enter Array1:")

5 arr1 = np.array([list(map(int, input().split())) for i in range(3)])

6

7 print("Enter Array2:")

8 arr2 = np.array([list(map(int, input().split())) for i in range(3)])

9

10 # Perform horizontal stacking (hstack)

Average time0.023 s22.76 ms

Maximum time0.026 s26.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 126 ms

Debug

Expected output

Actual output

Enter Array1:

Enter Array1:

1 2 3

1 2 3

4 5 6

4 5 6

7 8 9

7 8 9

Enter Array2:

Enter Array2:

4 5 6

4 5 6

Terminal

Test cases

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3.2.1. Numpy: Matrix Operations

23/53

The given code takes two  $3 \times 3$  matrices, `matrix_a`, and `matrix_b`, as input from the user and converts them into NumPy arrays.

**Task:**

You are required to compute and display the results of the following matrix operations:

1. **Addition** (`matrix_a + matrix_b`)
2. **Subtraction** (`matrix_a - matrix_b`)
3. **Element-wise Multiplication** (`matrix_a * matrix_b`)
4. **Matrix Multiplication** (`matrix_a . matrix_b`)
5. **Transpose of Matrix A**

**Input Format:**

- The user will input 3 rows for `matrix_a`, each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for `matrix_b`, each containing 3 integers separated by spaces.

**Output Format:**

The program should display the results of the operations in the following order:

1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

matrixOp...

Submit

Debugger

11 # Addition

12 print("Addition (A + B):")

13 print(matrix\_a + matrix\_b)

14 # Subtraction

15 print("Subtraction (A - B):")

16 print(matrix\_a - matrix\_b)

17 # Multiplication (element-wise)

18 print("Element-wise Multiplication (A \* B):")

19 print(matrix\_a \* matrix\_b)

20 # Matrix multiplication (dot product)

Average time 0.044 s 44.00 ms

Maximum time 0.088 s 88.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 88 ms Debug

Expected output

Enter Matrix A:

1 2 3

4 5 6

7 8 9

Enter Matrix B:

1 1 1

Actual output

Enter Matrix A:

1 2 3

4 5 6

7 8 9

Enter Matrix B:

1 1 1

Terminal Test cases

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3.1.1. Numpy Array Operations

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using `ndim`.
- The shape of the array using `shape`.
- The total number of elements in the array using `size`.

Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers, `r` and `c`, representing the number of rows and the number of columns.
- The next `r` lines contain `c` space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

Note: Use `reshape()` function to reshape the input array with the specified number of rows and columns.

Sample Test Cases

numpyarr...

1 import numpy as np

2 row, cols = map(int, input().split())

3 elements = []

4 for \_ in range(row):

5 elements.extend(map(int, input().split()))

6 arr = np.array(elements).reshape(row, cols)

7

8 print(arr)

9 print(arr.ndim)

10 print(arr.shape)

Average time

0.015 s

14.58 ms

Maximum time

0.041 s

41.00 ms

2 out of 2 shown test case(s) passed

10 out of 10 hidden test case(s) passed

Test case 1

Expected output

3 4

1 2 3 4

5 6 7 8

9 10 11 12

[[ 1 2 3 4]

[ 5 6 7 8]

Actual output

3 4

1 2 3 4

5 6 7 8

9 10 11 12

[[ 1 2 3 4]

[ 5 6 7 8]

Terminal

Test cases

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